DASHAN DONG

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Academic Experience

Postdoctoral Associate Oct 2023 - now

Prof. Ji-xin Cheng's Lab, ECE, Boston University *Boston, MA, USA*

Associate Research Fellow Jul 2021 - Oct 2023

Prof. Kebin Shi's Lab, School of Physics, Peking University *Beijing, China*

Postdoctoral Fellow Jul 2019 - Jun 2021

Prof. Kebin Shi's Lab, School of Physics, Peking University *Beijing, China*

Education

Peking University *Ph.D. in Optics* Advisor: Prof. Kebin Shi 2014 - 2019

Nankai University *BS in Physics* 2010 - 2014

Highlighted Publications

(1) *Xu Zhang#, Yonghao Chen, Shulei Zhou, Xiaoshuai Huang, Yujie Sun, Hong Yang,* ***Dashan Dong***✉*, Yue Wang*✉*, and Kebin Shi*✉. " Common-path dark-field full-field optical coherence tomography with asymmetric detection and spatial illumination " *Light: Science & Applications*, In press, (2026).

(2) **Dashan Dong#**, *Danchen Jia, Xinyan Teng, Jianpeng Ao, George Abu-Aqil, Biwen Gao, Meng Zhang, Qing Xia, Ji-Xin Cheng*✉. " Chem-SIM: SIM-resolved chemical imaging via wide-field mid-infrared photothermal modulation of fluorescence " *Nature Communications*, (2026).

(3) *Lige Liu#,* **Dashan Dong#**, *Zhiwei Long, Wanxue Wei, Chang Sun, Wei Liu, Xiaoshuai Huang, Liangyi Chen, Haizheng Zhong*✉*, and Kebin Shi*✉. " Single particle fluorescence imaging of perovskite nanocrystal crystallization for illustrating coupled nucleation-and-growth " *Nature Communications*, 16.5664, (2025).

(4) *Shuqi Mu#, Yingtong Shi, Yintong Song, Wei Liu, Wanxue Wei, Qihuang Gong,* **Dashan Dong**✉*, and Kebin Shi.* " Multislice computational model for birefringent scattering " *Optica*, 10.1, (2023).

(5) *Shuqi Mu#, Yu Yang, Juhao Li,* ***Dashan Dong***✉*, Ruijun Lan*✉*, Kebin Shi.* " Three dimension refractive index characterization for photonic waveguides " *Journal of Lightwave Technology*, 40.8, (2022).

(6) **Dashan Dong#**, *Xiaoshuai Huang#, Liuju Li#, Heng Mao, Yanquan Mo, Guangyi Zhang, Zhe Zhang, Jaiyu Shen, Wei Liu, Zeming Wu, Guanghui Liu, Yanmei Liu, Hong Yang, Qihuang Gong, Kebin Shi*✉*, and Liangyi Chen*✉. " Super-resolution fluorescence-assisted diffraction computational tomography reveals the three-dimensional landscape of the cellular organelle interactome " *Light: Science & Applications*, 9.11, (2020).

Other Publications (Peer-Reviewed)

(1) *Ziyu Li#, Yue Wang#*✉*, Chendi Shao, Jian Xu,* **Dashan Dong**, *Shulei Zhou, Xiaoshuai Huang, Hong Yang, Yujie Sun, Kebin Shi*✉. " Depth-Robust Super-Resolution Nanofabrication for 3D Mesoscopic Photonic Structures " *Nature Nanotechnology*, In press,(2026).

(2) *Liuju Li#, Miao He#, Bulidierxin Tuerhanbayi ,* ***Dashan Dong***, *Wenqing Lu, Yuqing Zhang, Wang Yao, Liang Yang, Yujun Liu, Linglong Li, Ziying Luo, Shuai Yao, Xinyue Chen, Ming Shi, Xuefei Zhang, Tongsheng Chen, Yangming Wang, Tingting Li,*

Liying Yan, Hyunsoo Je, Jie Qiao, Kebin Shi, Xiaoshuai Huang✉*, and Liangyi Chen*✉. " Label-free super-resolution imaging reveals multicompartment glycogen condensates as metabolic regulators of pluripotency in embryonic stem cells ." *Nature Communications*, In press,(2026).

(3) *Jianpeng Ao*#*, Jiaze Yin*#*, Haonan Lin, Guangrui Ding, Youchen Guan, Marzia Savini, Bethany Weinberg,* **Dashan Dong**, *Qing Xia, Zhongyue Guo, Bowen Liu, Biwen Gao, Ji-Xin Cheng*✉*, and Meng C Wang*✉. " FILM: mapping organellar metabolism by mid-infrared photothermal-modulated fluorescence ." *Nature Methods*, 23.1196-1206, (2026).

(4) *Danchen Jia*#*,* **Dashan Dong**#*, Tongyu Li, Haonan Zong, Jiabei Zhu, Xinyan Teng, Lei Tian*✉*, Ji-Xin Cheng*✉. " Mid-Infrared Photothermal Relaxation Intensity Diffraction Tomography for Video-rate Volumetric Chemical Imaging ." *Optics Express*, 34.9, (2026).

(5) *Kai Zhang*#*, Shuqi Mu*#*,* **Dashan Dong**, *Qihuang Gong, Yue Wang, and Kebin Shi*✉. " Full-component tomographic reconstruction of a dielectric tensor using a multislice computational model ." *Advanced Imaging*, 3.3, (2026).

(6) *Tongyu Li*#*, Yi Shen,* **Dashan Dong**, *Danchen Jia, Jianpeng Ao, Ji-Xin Cheng, Lei Tian*✉. " Transfer-function approach to substrate-enhanced diffraction tomography ." *Optica*, 13.4, (2026).

(7) *George Abu-Aqil*#*,* **Dashan Dong**#*, Jiaze Yin*#*, Jianpeng Ao, Hongjian He, Guangrui Ding, Qing Xia, Ji-Xin Cheng*✉. " Bond-Selective Imaging via Vibrational Relaxation Encoded Fluorescence ." *The Journal of Physical Chemistry Letters*, 16.48, (2025).

(8) *Jia Danchen*#*, Huang Steven*#*, Tulegenov Dias,* **Dashan Dong**, *Shvets Gennady*✉*, Cheng Ji-Xin*✉. " Metasurface-enhanced infrared photothermal microscopy towards ultrasensitive chemical analysis ." *Advanced Photonics*, 7.5, (2025).

(9) *Qian Chen*#*, Wen Gou*#*, Wenqing Lu*#*, Jie Li, Yuhong Wei, Haoyu Li, Chengyu Wang, Wei You, Zhengqian Li,* **Dashan Dong**, *Xiuli Bi, Bin Xiao*✉*, Liangyi Chen*✉*, Kebin Shi*✉*, Junchao Fan*✉*, Xiaoshuai Huang*✉. " Fast, three-dimensional, live-cell super-resolution imaging with multiplane structured illumination microscopy ." *Nature Photonics*, 19.567-576, (2025).

(10) *Jiaze Yin*#*, Christian Pfluegl, Chu C. Teng, Rylie Bolarinho, Guo Chen, Xinrui Gong,* **Dashan Dong**, *Daryoosh Vakhshoori, Ji-Xin Cheng*✉. " Mid-infrared Energy Deposition Spectroscopy ." *Physical Review Letters*, 134.9, (2025).

(11) *Yue Wang*#*, Jingrui Gong, Ning Xu, Shaohui Yan,* **Dashan Dong**, *Kebin Shi*✉. " Large Field of View and Isotropic Light Sheet Microscopy with Aberration-Free Tunable Foci ." *Laser & Photonics Reviews*, 19.3, (2025).

(12) *Wenkai Yang*#*, Zijian Wang, Jian Xu,* **Dashan Dong**, *Guiyuan Cao, Han Lin, Baohua Jia, Lige Liu*✉*, Kebin Shi*✉. " Ultra-compact computational spectroscopy with a detour-phased planar lens ." *Light: Advanced Manufacturing*, 5.44, (2024).

(13) *Ziheng Ji*#*, Wentao Yu,* **Dashan Dong**, *Hong Yang, Kaihui Liu, Yun-Feng Xiao, Qihuang Gong, Qinghai Song*✉*, Kebin Shi*✉. " High spatial resolution collinear chiral sum-frequency generation microscopy ." *Advanced Photonics Nexus*, 3.2, (2024).

(14) *Yue Wang*#*,* **Dashan Dong**, *Wenkai Yang, Renxi He, Ming Lei, and Kebin Shi*✉. " Reflective ultrathin light-sheet microscopy with isotropic 3D resolutions ." *Photonics Research*, 12.2, (2024).

(15) *Peng Liu*#*, Jing Shi*#*, Danli Sheng*#*, Wenqing Lu, Jie Guo, Lei Gao, Xiaoqing Wang, Shaofeng Wu, Yanwen Feng,* **Dashan Dong**, *Xiaoshuai Huang*✉*, and Hongyun Tang*✉. " Mitopherogenesis, a form of mitochondria-specific ectocytosis, regulates sperm mitochondrial quantity and fertility ." *Nature Cell Biology*, 25.1625-1636, (2023).

(16) 魏婉雪 #, 何苗, 徐坚, 董大山✉. " 一种大视野快速光学衍射层析成像技术 ." 中国激光, 50.3, (2023).
Wanxue Wei#*, Miao He, Jian Xu,* **Dashan Dong**✉. " Large field-of-view fast optical diffractive tomographic microscopy ." *Chinese Journal of Lasers*, 50.3, (2023).

(17) *Wenkai Yang*#*, Lige Liu*✉*,* **Dashan Dong**, *Yunan Gao, Hong Yang, Qihuang Gong, and Kebin Shi*✉. " *In situ* three-dimensional observation of perovskite crystallization revealed by two-photon fluorescence imaging ." *Advanced Optical Materials*, 10.13, (2022).

(18) *Wenkai Yang*#*, Lige Liu*#*,* **Dashan Dong**, *Xin Zhang, Han Lin, Yunkun Wang, Hong Yang, Yunan Gao, Haizheng Zhong, Baohua Jia, and Kebin Shi*✉. " Detour-phased perovskite ultrathin planar lens using direct femtosecond laser writing ."

Photonics Research, 10.12, (2022).

(19) *Runfeng Li#, Ruijun Lan✉*, **Dashan Dong**, *Hong Yang, and Kebin Shi✉*. " Bessel beam coherent anti-Stokes Raman scattering spectroscopy for turbulent flow diagnosis ." *Applied Spectroscopy*, 77.3 (2022).

(20) 穆书奇 #, 董大山, 施可彬✉. " 无标记光学成像技术 ." 激光与光电子学进展, 59.12, (2022).
*Shuqi Mu#, **Dashan Dong**, Kebin Shi✉*. " Label-free optical imaging technology ." *Laser & Optoelectronics Progress*, 59.12, (2022).

(21) 李润丰 #, 董大山, 施可彬✉. " 光场调控在相干拉曼散射光谱与成像中的应用 (特邀) ." 光子学报, 51.1, (2022).
Runfeng Li#, **Dashan Dong**, *Kebin Shi✉*. " Coherent Raman scattering spectroscopy and microscopy based on optical field engineering (Invited) ." *Acta Photonica Sinica*, 51.1, (2022).

(22) *Wei Liu#*, **Dashan Dong**, *Hong Yang, Qihuang Gong, and Kebin Shi✉*. " Robust and high-speed rotation control in optical tweezers by using polarization synthesis based on heterodyne interference ." *Opto-Electronic Advances*, 3.8, (2020).

(23) *Shuanglong Liu#, Bowen Sheng, Xinqiang Wang✉*, **Dashan Dong**, *Ping Wang, Zhaoying Chen, Tao Wang, Xin Rong, Duo Li, Liuyun Yang, Shangfeng Liu, Mo Li, Jian Zhang, Weikun Ge, Kebin Shi, Yuzhen Tong, Bo Shen.* " Molecular beam epitaxy of single-crystalline aluminum film for low threshold ultraviolet plasmonic nanolasers ." *Applied Physics Letters*, 122.23, (2018).

(24) *Wentao Yu#, Ziheng Ji*, **Dashan Dong**, *Xusan Yang, Yunfeng Xiao, Qihuang Gong, Peng Xi✉, and Kebin Shi✉*. " Superresolution deep imaging with hollow Bessel beam STED microscopy ." *Laser & Photonics Reviews*, 10.1, (2016).

(25) *Yonggang Lv#, Ziheng Ji*, **Dashan Dong**, *Kebin Shi✉, and Qihuang Gong*. " Wide-field vibrational phase contrast imaging based on coherent anti-Stokes Raman scattering holography ." *Chinese Physics Letters*, 32.7, (2015).

Other Publications (without Peer-Review)

(1) **Dashan Dong#**, *Kebin Shi✉*. " Solving the missing cone problem by deep learning ." *Advanced Photonics*, 2.2, (2020).

(2) *Bin Tsai#, Wei Liu*, **Dashan Dong**, *Kebin Shi, Liangyi Chen, Ning Gao✉*. " Phase separation of Mer2 organizes the meiotic loop-axis structure of chromatin during meiosis I ." *bioRXiv*, (2020).

Conference Abstracts

(1) **Dashan Dong** *and Ji-Xin Cheng*. " Super-resolution Chemical Imaging via Structured Illumination Fluorescence-Detected Mid-Infrared Photothermal Microscopy ." *Novel Techniques in Microscopy*, Optica Biophotonics Congress NTu3C.3 (2025).

(2) *Wanxue Wei*, **Dashan Dong**, *Kebin Shi.* " Fast optical diffraction tomography microscopy with large field of view and lossless pupil beam combination ." *Ultrafast Nonlinear Imaging and Spectroscopy XII*, Proc. SPIE 1313910, (2024).

(3) **Dashan Dong**, *Xiaoshuai Huang, Liuju Li, Kebin Shi, and Liangyi Chen.* " Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome ." *Advanced Optical Imaging Technologies III*, SPIE/COS Photonics Asia 115490G, (2020).

(4) **Dashan Dong**, *Xiaoshuai Huang, Liuju Li, Kebin Shi, and Liangyi Chen.* " Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome ." *Digital Holography and Three-Dimensional Imaging 2020*, Imaging and Applied Optics Congress, Optica Publishing Group HF1G.6, (2020).

(5) **Dashan Dong**, *Yanhui Cai, Ziheng ji, Hong Yang, Qihuang Gong, and Kebin Shi.* " Tomographic Diffractive Microscopy for Better 3D Imaging ." *14th International Conference on Photonics and Imaging in Biology and Medicine*, Optica Publishing Group W3A.43, (2017).

(6) *Kebin Shi*, **Dashan Dong**, *Yanhui Cai, Wei Liu, Chendi Shao.* " High resolution nonlinear imaging based on optical field engineering ." *Ultrafast Nonlinear Imaging and Spectroscopy V*, Proc. SPIE 1038000Z, (2017).

Patents

(1) *System and Method for Super-Resolution Chemical Imaging Via Photothermal*
International Patent PCT/US26/24003, (2026)

(2) Dual-modality microscopic imaging system and method — International Patent PCT/CN2021/071393, (2020)

(3) Off-axis holographic beam combining device and method based on missing reflector
CN Patent 2023110715739, (2023)

(4) Dual-modality microscopic imaging system and imaging method — CN Patent ZL202110946936.3, (2022)

(5) Reflective axial light sheet fluorescence microscopy imaging device and method based on microprism
CN Patent ZL 202110817118.3, (2022)

(6) Bessel CARS-based turbulence spectral measurement system and its detection method
CN Patent ZL202110667298.1, (2022)

(7) Dual-modality microscopic imaging system and method — CN Patent ZL202010059510.1, (2022)

(8) Device and method for generating dynamic cylindrical vector light field based on optical heterodyne interferometry
CN Patent ZL202010493775.2, (2021)

(9) GB-STED based deep-layer super-resolution laser direct writing system and implementation method thereof
CN Patent ZL202010069141.4, (2020)

Received Grants

[Co-PI] Young Scientists Sub-Project (Label-free imaging and big-data machine learning) — 2022
Super-resolution 3D landscape imaging and resolving technology of living cell — CNY 2,340,000
National Key Research and Development Program of China

[Co-PI] Clinical Medicine Plus X - Young Scholars Project — 2022
Exploring the role of lipid droplets in egg asymmetric division based on label-free imaging — CNY 100,000
Peking University, the Fundamental Research Funds for the Central Universities

[PI] Young Scientist Fund (12004013) — 2021
Reflective optical diffraction tomography for live cell imaging and its application — CNY 240,000
National Natural Science Foundation of China

[PI] No.68 General Fund (2020M680220) — 2021
Super-resolution fluorescence-assisted optical diffraction tomography and its application — CNY 80,000
China Postdoctoral Science Foundation

Conference Presentations

[Oral] Optica Biophotonics Congress 2025 — Apr 2025
Super-Resolution Chemical Imaging via Structured Illumination Fluorescence-Detected Mid-Infrared Photothermal Microscopy — *San Diego*

[Poster] Chemical Imaging 2023, Gordon Research Conference — Aug 2023
Optical Diffraction Tomography Reveals the 3D Landscape of Living Cells — *Boston, USA*

[Invited] Advanced Imaging and Information Processing Conference (AIIP 2023) — Jul 2023
New Technologies for Bio-Optical Imaging with Low Phototoxicity — *Jinggangshan, China*

[Invited] 第三届全国光子技术论坛 — Apr 2023
High-Resolution Imaging Based on Light Field Modulation and Applications — *Guangzhou, China*

[Invited] International Computational Imaging Conference (CSOE-CITA 2022) — Oct 2022

Live Cell Landscape Imaging Based on Optical Diffraction Tomographic Microscopy — *Shanghai, China*

[Invited] 5th Optics Young Scientist Summit (OYSS2022) — Sep 2022
High-Resolution Imaging Based on Light Field Modulation and Applications — *Fuzhou, China*

[Invited] 20th National Symposium on Basic Optics and Optical Physics — Sep 2021
Live Cell Optical Diffraction Tomographic Imaging Based on 4D Spatiotemporal Continuity — *Yanji, China*

[Invited] The Chinese Optical Society (COS) Academic Conference 2021 — Sep 2021
Optical Diffraction Tomographic Microscopy Based on 4D Spatiotemporal Continuity — *Shenzhen, China*

[Invited] 4th Optics Young Scientist Summit (OYSS2020) — Dec 2020
Optical Diffraction Tomographic Microscopy and Its Applications — *Ningbo, China*

[Oral] 第二届全国光子技术论坛 — Nov 2020
Optical Diffraction Tomographic Microscopy and Its Applications — *Guangzhou, China*

[Invited] Chinese Electron Microscopy Society Academic Conference 2020 — Nov 2020
Live Cell Landscape Imaging Based on Optical Diffraction Tomographic Microscopy — *Chendu, China*

[Oral] The 12th National Academic Forum on Optical Youth — Nov 2020
Optical Diffraction Tomographic Microscopy and Its Applications — *Baoding, China*

[Oral] SPIE/COS Photonics Asia 2020 — Oct 2020
Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome — *Webinar*

[Oral] Digital Holography and Three-Dimensional Imaging, Imaging and Applied Optics Congress — Jun 2020
Super-Resolution Fluorescence Assisted Diffraction Computational Tomography Reveals the Three-Dimensional Landscape of Cellular Organelle Interactome — *Webinar*

[Poster] International Conference on Photonics and Imaging in Biology and Medicine — Sep 2017
Tomographic Diffractive Microscopy for Better 3D Imaging — *Suzhou, China*

[Poster] SPIE Optics + Photonics 2017 — Aug 2017
Tomographic Diffractive Microscopy for Better 3D Imaging — *San Diego, USA*

[Oral] Chinese Physical Society (CPS) Fall Meeting — Aug 2016
Optical Diffraction Tomographic Microscopy — *Beijing, China*

Services

[Reviewer]	Photonics Research, Optics Letters, Optics Express, Advanced Science, APL Photonics, Chinese Journal of Lasers, Acta Optica Sinica	
[Program Committee]	Topic 9: Biomedicine and Computational Imaging, International Computational Imaging Conference (CSOE-CITA)	2022, 2023
[Youth Editor]	Chinese Journal of Lasers	2022, 2023
[Volunteer]	SPIE/COS Photonics Asia	2016, 2022
[President]	Peking Univ. Club, SPIE Student Chapter	2017
[Vice President]	Peking Univ. Club, SPIE Student Chapter	2016

Theses

[Doctoral Thesis] *Research on Optical Diffraction Tomographic Microscopy* — Supervisor: Prof. Kebin Shi, Peking University

[Bachelor Thesis] *Characteristics of Novel Vector Field* — Supervisor: Prof. Yongnan Li, Nankai University

Skills

[Optics] **Expert** in *Computational Imaging, Nonlinear Optics, Holography/Digital Holography, Microscopy, Spectroscopy*
Skilled in *Designing, Building, Aligning, and Optimizing Complex Optical Imaging Systems*
Proficient in *Live-Cell Imaging, Label-free and Fluorescence Imaging, Multimodal and Super-resolution Microscopy*

[Programming] MATLAB, C/C++, Labview, Python, Bash, LaTeX
- Proficient in **Instrument Control**, with expertise in *Imaging/Signal Acquisition* and *Embedded System*
- Skilled in **Image/Signal Processing**, with experience in *Computer Vision, Data Visualization*
- Experienced in **High Performance Computing** with *Parallel Programming* on *GPU and Linux-Clusters*

[Softwares] Fiji/ImageJ, Mathematica, OriginPro, Adobe AI/PS, Zemax, COMSOL

Honors & Awards

[2022] Gan Zizhao Scholarship for Outstanding Postdoctoral Fellows *Peking University*

[2022] Important Advance for Institute of Modern Optics *Peking University*

[2017] May 4th Scolarship *Peking University*

[2017] Individual Honor for Research Excellence *Peking University*

[2013] He-Zhang Encouragement Scholarship
Foundation for the Development of Science and Technology Museums in China

[2012] National Encouragement Scholarship *Nankai University*

[2012] Merit Student *Nankai University*

[2011] Outstanding Student Scholarship in Fundamental Science *Nankai University*